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Mass balance modelling of potassium losses from grazed dairy pasture

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Abstract A simple mass balance model was used to estimate the annual losses of potassium (K) from three dairy farms. The results showed that while the amount lost from each farm varied, 74–92% of the losses were directly caused by the grazing dairy cow through product removal from the farm, excreta transfer to unproductive areas, and preferential flow of urine K through the soil to beyond the effective plant rooting depth. A net loss of K leads to reductions in soil K, and for one of the farms (on an organic soil with a low total soil K content) it was possible to show that the predictions of K losses made by the model were comparable with actual changes in soil K status. The K losses per stock unit (su) estimated by this model for the three farms ranged from 3.4 to 4.7 kg K/su and contrasted with a loss of 4.0 kg K/su in the model used by the Ministry of Agriculture and Fisheries (MAF) for predicting K fertiliser requirements. Refinements to the MAF model are suggested for calculating the animal loss factor.

Keywords potassium; model; K cycle; nutrient cycling; potassium losses; dairy cattle; fertiliser requirements

INTRODUCTION

The application of potassium (K) fertiliser to New Zealand dairy pastures is important for maintaining pasture growth and milk production (During 1984). The amount of fertiliser required to maintain productivity at a desirable level can be predicted by calculating the net loss of K from the farming system (Campkin 1985). A simple model capable of estimating the net loss of K from any dairy farming system can be constructed from a mass balance of losses from and gains to the K cycle of grazed dairy pasture.

Losses from the farm occur through the removal of K in animal products (milk, calves, and culled cows), the transfer of excretal K to the unproductive areas of the farm (e.g., dairy shed, raceways, and feedpads), removal of hay and silage for sale and leaching below the plant rooting depth. Losses of K as a result of leaching can occur in two ways. Following urination, a proportion of the urine may be leached because of preferential flow through the soil macropores (Williams et al. 1990). K can also be leached in percolating drainage water following rain or irrigation. Gains of K to the farm are only normally through fertiliser application but are sometimes considered to occur through release of K from reserves (e.g., micaceous clay minerals) in the soil.

In New Zealand, a simple model is used by the Ministry of Agriculture and Fisheries (MAF) to calculate losses of K from farms and therefore predict fertiliser requirements (Cornforth & Sinclair 1984). Despite the widespread use of this model no information has been published on the accuracy of the predicted losses in the field. The main reason for this lack of appraisal has been the scarcity of suitable data from which to measure or estimate the losses for a given system. There is, however, some evidence which suggests that the model overestimates some of the losses of K which occur from grazed dairy pasture (Williams & Roberts 1988). Better quantification of the losses of K from dairy pastures is necessary to improve predictions of K fertiliser requirements.

In this paper, a simple model (subsequently referred to as the K Loss Model) has been used to calculate the annual K losses for grazed pastures on three dairy farms. In theory, net K losses over many years will lead to measurable decreases in soil K. For the three farms, an attempt was made to compare the predicted change in soil K created by these losses (or gains) with actual changes in soil K. A comparison was also made between the K Loss Model and the MAF model for predicting K losses.

MATERIALS AND METHODS

Sites

The farms studied represent three major dairying regions in New Zealand (Waikato, Manawatu, and Taranaki). They are sited on different soil types, and provide a range of animal management and production systems. Table 1 gives details of the farms. Information on the losses of K which occur from the Taranaki farm via animal production and transfer has been published by Williams & Roberts (1988).

K Loss Model

The K Loss Model estimates the amounts of K lost annually in animal products, in transfer to unproductive areas and during preferential flow of urine. The details of the equations involved are given in Appendix 1. The losses were calculated on a per cow basis and in this paper have been expressed as a loss/ha and as a loss/stock unit (su) (where one (su) has an intake of 550 kg dry matter (DM)/year). Basing the loss on a su rather than on a per cow basis allows for differences in pasture intake that

occur between cows of different liveweight and productivity (Cornforth & Sinclair 1984).

Estimates of the amounts of K leached in the drainage water from rain events have been calculated on a per ha basis from the amount of drainage/ha and the K concentration in this drainage.

The boundary condition for the K loss by leaching has been taken to be a soil depth of 15 cm, because the zone of maximum root activity for K uptake from urine patches by pasture plants in a central yellow-grey earth has been shown to occur above this depth (Williams et al. 1989). In the absence of other information, we have assumed that the recovery of urine K from soil depths greater than 15 cm is negligible in the soils in this study. The soils are all from areas which receive an annual rainfall greater than 1000 mm and include the central yellow-grey earth soil. Thus any K which is leached below 15 cm is considered to be lost from the grazed pasture system.

Parameters in the model

Losses from animal products

The amount of K lost from the farm via animal products is the sum of the losses that occur in milk and calf production and culled cows. These three components were calculated separately (Equations 1, 2, and 3, Appendix 1) based on data of milkfat production, mean annual milkfat content, mean cow liveweight, and length of time each cow spends in the herd.

Losses from transfer of excretal potassium to unproductive areas

The amount of K that was lost by transfer to unproductive areas (Equation 4, Appendix 1) is based on the amount of time that the dairy cows spend in these areas (Williams & Roberts 1988). For instance, if the cows are milked for 260 days of the year, and during this period they are off the paddock for an average of 2 h/day, then they spend 6% of the year in unproductive areas and 6% of the excreta is deposited there.

The total amount of K excreted by the cows is the difference between the amount of K ingested and the amount of K lost in animal products. Since measurements of cow intake are generally not available on dairy farms, the amount of K ingested was calculated from the milkfat production and the conversion factor of 25 kg of DM for each kg of milkfat produced (Bryant 1982). The amount of K ingested was then estimated by assuming that the

Table 1 Background information on sites.

	Waikato	Manawatu	Taranaki
Annual rainfall (mm)	1200	1000	1000
N.Z. soil group	Organic	Yellow-grey earth	Yellow-brown loam
Soil taxonomic classification	Typic Medisaprist	Typic Fragiaqualf	Entic Dystrandept
Mean stocking rate (cows/ha)	2.7	3	3.5
(su/ha)	17	21	25
Mean milkfat production (kg/ha)	340	450	640
Mean fertiliser application rate (kg K/ha per year)	74	0	70

DM ingested had K contents of 3.2, 2.0, and 3.5% for the Waikato, Manawatu, and Taranaki farms respectively. These average herbage K concentrations are based on actual measurements on the farms and data from Metson & Saunders (1978).

Preferential urine loss

Of the excretal K deposited on the pasture, 90% is considered to be in the form of urine (Hutton et al. 1967). Measurements carried out on the Waikato, Manawatu, and Taranaki farms showed that 13, 30, and 12% of the urine K would be expected to be lost preferentially (Williams et al. 1990). The amounts of K that were lost via this mechanism were calculated using Equation 5 in Appendix 1.

Leaching losses from rainfall

The amount of K leached/year from the rainfall events was calculated from the amount of drainage/year (estimated from the sum of the amounts of rainfall which exceeded evapotranspiration each month with an allowance for soil water storage) and K concentrations of 4 (Waikato and Taranaki farms; Campkin 1985) and 3 µg/ml (Manawatu farm; Smith et al. 1984).

Although the leaching losses have been partitioned into two categories (rainfall and preferential flow of urine) and each has been calculated separately, some of the K in the drainage water may have originated from preferential flow of urine through the soil into the drains. Conversely, not all of the K that is leached beyond 15 cm makes its way into the drainage water. Thus calculating these losses separately may result in an overestimation of the amount of urine K lost preferentially and an underestimation of the amount of K leached from rain events.

Table 2 Mean annual losses of potassium calculated by the K Loss Model.

	Waikato		Manawatu		Taranaki	
	kg/ha	kg/su	kg/ha	kg/su	kg/ha	kg/su
Animal products	10	0.6	13	0.6	17	0.7
Transfer to unproductive areas of farm	16	1.0	39	1.8	24	1.0
Preferential loss of urine	29	1.8	48	2.3	54	2.1
Drainage water	19	1.1	9	0.4	11	0.4
Total loss	74	4.5	109	5.1	106	4.2

RESULTS AND DISCUSSION

Losses of potassium

Table 2 shows the estimations of K losses for the three farms. The largest loss of K (29–54 kg K/ha or 1.8–2.3 kg K/su) occurred through the preferential flow of urine K, whereas the smallest losses occurred via animal products (10–17 kg K/ha or 0.6–0.7 kg K/su) and leaching in drainage water (9–19 kg K/ha or 0.4–1.1 kg K/su).

The losses in Table 2 highlight the role of the dairy cows in causing K losses from grazed pasture. The losses of K via animal products, transfer to unproductive areas, and through the preferential loss of urine can be directly attributed to the grazing dairy cows and amounted to 74–92% of the calculated total losses.

There was a considerable variation in the size of the losses among the three farms, with the Waikato farm having the smallest total loss. One factor which controls the size of the K losses is the amount of K that is cycling through the grazed pasture. Since the Waikato farm had the lowest animal production and stocking rate compared with the other farms (Table 1), there was a smaller amount of K being cycled and consequently the losses of K/ha were smaller. A comparison of the K losses/su also indicates a variation between the three farms. The greatest variability was the transfer to unproductive areas of the farm, which is influenced by farm management. Since the cows on the Manawatu farm spend an average of 25 days on a feed pad/year whereas the cows on the other farms were not kept on a feedpad at all, the transfer loss on the Manawatu farm was the highest (Table 2).

Validation of model

The total amount of K lost on the Manawatu and Taranaki farms (109 and 106 kg K/ha; Table 2) exceeded the amount of fertiliser that was applied annually (0 and 70 kg K/ha; Table 1). Therefore it could be expected that there would be a decline in the soil K concentration on these two farms.

On the Taranaki farm, soil K concentration has been measured annually by the MAF Quick Test (Cornforth 1980). This test extracts solution K and c. 70% of exchangeable K. For a soil with a bulk density of 0.8 Mg/m³ one Quick Test unit is equivalent to 0.07 me K/100 g. Over the 8-year period of this study, there was no indication that the Quick Test extractable K content of the soil was declining (Williams & Roberts 1988) even though the total accumulated loss could have resulted in a

drop of 10 Quick Test units (from 12 to 2) in the 0–7.5 cm sampling depth (assuming that pasture plants remove K equally from the 0–15 cm depth of soil). The fact that a change of this magnitude was not observed indicates that the soil K pool as measured by the Quick Test was being replenished by release of K from other sources in the soil and/or that actual K losses were smaller than predicted.

Despite annual losses of 109 kg K/ha on the Manawatu farm with no K fertiliser input, pasture K analyses, small plot observation, and mowing trials over the last 5 years have failed to provide any conclusive evidence that the farm is responsive to K fertiliser (P. E. H. Gregg unpubl. data). It must be assumed therefore, that the rate of release of soil K from the exchangeable and nonexchangeable sources in the soil is sufficient to meet this loss. This observation is not consistent with the MAF model which calculates the rate of K supply by the soil as 40 kg K/ha per year (Campkin 1985).

The large amounts of total K present in the 0–15 cm depth of soil in the Taranaki and Manawatu farms (c. 17 000 kg K/ha; Metson 1980) and the continuous replenishment of exchangeable K from the extremely large pool of nonexchangeable and structural forms of soil K, means that changes in the quantity of K in the soil caused by losses of K over short time periods (e.g., 1–8 years) are less than 1–2% of total soil K. Sampling errors associated with total soil K variability in the field and analytical errors are often 18% of the mean (unpubl. data), thus changes in soil K of a few hundred kg are difficult to measure on these farms. The Waikato farm is on an organic soil which has a very low native soil K content (c. 300 kg K/ha) and a virtual absence of soil minerals that release nonexchangeable K. These features mean that it is possible to

compare predicted losses and actual changes in soil K for this farm.

Potassium losses for the 30 years since the Waikato farm was developed have been calculated (Table 3) based on the data that were available and best estimates of the data that were not (e.g., early animal production data and herbage K concentrations). The amount of K gained by the farm over the 30-year period via fertiliser applications has also been calculated (Table 3).

Measurements of the actual changes in soil K over the 30-year period were made by comparing the K status of a site on the dairy farm with the closest adjacent area of undeveloped Moanatuatua swamp (1 km away). For the purpose of comparing the K status of the developed and undeveloped sites, it was assumed that these sites were pedologically similar, and that over the last 30 years the undeveloped site remained in steady state equilibrium.

During development, the surface of the peat bog subsided because of shrinkage as the peat dried out and the organic matter decomposed (Hupkens van der Elst 1980). To measure the K content of the corresponding depth of undeveloped peat that subsided to form the 0–15 cm depth of developed soil, it was necessary to estimate the rate of subsidence for this site. Soil samples were collected down to a depth of 60 cm (water table depth) in 10 cm increments at the undeveloped site, and at the developed site samples were collected to the 30 cm soil depth in 5 cm increments. Samples were collected from six soil profiles chosen randomly at each site. The amount of mineral matter in each soil depth in the developed and undeveloped sites was measured using the weight loss as a result of ignition. The soil samples were also ashed, digested with acid (Beckwith & Little 1963), and the K and iron (Fe) contents of the extract measured by atomic spectroscopy.

At both sites, soil bulk density did not change significantly with changes in soil depth. There was, however, a difference in bulk density between the sites, viz., 0.55 Mg/m³ (SEM = 0.01) in the 0–30 cm depth of developed soil and 0.12 Mg/m³ (SEM = 0.01) for the 0–60 cm depth of undeveloped soil.

There was a large difference in the amount of Fe present in the soil between the two sites. The increased amount of Fe in the 0–15 cm depth of soil at the developed site may have originated from subsidence of a certain original depth of undeveloped peat. By summing the Fe content of

Table 3 Total losses and gains of potassium over a 30-year period for the Waikato site.

Parameter	kg K/ha
Losses of K	
Animal products	312
Transfer to unproductive areas of farm	488
Preferential movement of urine	870
Drainage water	570
Total loss	<u>2240</u>
Gains of K	
Fertiliser	2226
Predicted net change in soil K	-14
Measured gain in soil K	<280

each soil depth in the undeveloped peat until the Fe content was equivalent to the Fe content in the 0–15 cm depth of developed peat, it can be calculated that the subsidence rate was greater than 4 m per 30 years. A similar calculation based on the weight loss on ignition data resulted in a subsidence rate of a similar magnitude. Measurements on the annual rate of subsidence of the Waikato peatlands have shown that the rate can be as high as 20 cm/year in the first year of development but then it declines to 2 cm/year (Thompson 1980). Therefore it is unlikely that the developed peat has subsided at a rate of greater than 4 m per 30 years. A more likely explanation is that the developed peat contained a higher mineral content initially compared with the undeveloped site. The different mineral content of the two sites may have originated through different patterns of deposition of Fe and K from flooding and volcanic ash showers (Green & Lowe 1985). This suggests that the assumption made earlier that the undeveloped and developed sites were initially pedologically similar, is incorrect.

In the 0–15 cm depth of soil on the developed site there were 280 kg K/ha (SEM = 56). The amount of K in the undeveloped peat was considerably smaller with 48 kg K/ha (SEM = 5) in the 0–60 cm depth. On the basis that the two sites were initially dissimilar, the actual change in K content of the soil to 15 cm depth can only be estimated at being less than or equal to 280 kg K/ha. The undeveloped site was the only one available in the area and no better estimate of the change in soil K status can be made.

When all possible fates for fertiliser K are considered (animal products, transfer to unproductive areas, and leaching losses) a balance sheet of the estimated losses and gains of K for 30 years on the farm shows that the estimated net change in soil K was –14 kg K/ha (Table 3). Therefore, it was expected that there would be little change in the soil K status over the 30 years.

The best estimate that could be made of the measured change in soil K showed a gain of less than 280 kg K/ha. Obviously there is a difference between the value for the predicted and measured change in soil K (–14 cf. <280 kg K/ha). This difference is small compared with the total amount of K that has either been lost from the farm (2240 kg K/ha) or added to the farm in fertiliser (2226 kg K/ha) over the 30 years. Therefore, it can be considered that the parameters used in the K Loss Model provided reasonable estimates of the K losses and gains for this farm.

Reducing potassium losses

Of the losses caused by dairy cows, the greatest loss (c. 50%) occurs through the preferential flow of urine K to soil depths beyond the effective plant rooting zone (Table 2). There are few if any practical ways of reducing this loss (Williams et al. 1990).

The second major loss relating to the animal is the amount of K that is transferred to unproductive areas such as the raceways, dairy shed, and feedpad. This loss of K could be reduced by minimising the time that cows spend in these areas through ensuring an efficient layout of raceways and gateways leading to and from the dairy shed (which minimises the time the cows spend standing off the paddock before and after milking) and better drainage of land to reduce the need for a feedpad (Williams et al. 1988). The collection and return of effluent from the dairy shed and feedpad to a large proportion of the farm will also help to reduce losses.

The quantity of K lost via dairy cow excreta is a function of the amount of K ingested, therefore the more K ingested the greater are the losses of K. It may be possible to reduce the amount of K ingested by reducing K fertiliser inputs or using split fertiliser applications in situations where herbage K concentration is above the critical level for pasture growth.

Model comparison

The MAF model (Cornforth & Sinclair 1984) was used to calculate the losses of K from the three farms in this study (Table 4). A comparison of the predictions made by this model and the K Loss Model (Table 4 cf. Table 2) reveal several differences. The MAF model predicted that losses of K because of transfer of excretal K to unproductive areas were much higher than those calculated by the K Loss Model. The reason for the difference between the values is that in the MAF model the assumption is made that dairy cows return 18% of their excreta to unproductive areas (Campkin 1985), whereas on the farms used in this study the cows returned less than 10% to unproductive areas even on the Manawatu farm where the cows spent several days of the year on a feedpad. This difference in excretal K return may result in the MAF model overestimating fertiliser requirements by 28–57 kg K/ha per year.

Whereas the MAF model may overestimate the amount of K lost through transfer, it underestimated the amount of K lost via leaching by not allowing for the loss that occurs during preferential flow of

Table 4 Losses of potassium calculated by the MAF model (kg K/ha).

	Waikato	Manawatu	Taranaki
Animal products	14	17	20
Transfer to unproductive areas of farm	53	67	81
Preferential loss of urine	0	0	0
Drainage water	11	3	19
Total loss	78	87	120

urine through the soil profile. The present study has shown this to be a major mechanism for K loss accounting for 50% of the K losses from the cycle (equivalent to 29–54 kg K/ha per year; Table 2). The MAF model uses two K loss values (Cornforth & Sinclair 1984), one for all dairy farms on flat topography (4 kg K/su) and another for farms on hill country (4.5 kg K/su). Presumably the higher loss on hill country is to allow for some transfer of excretal K to campsites using similar ideas to those developed from research with sheep (Gillingham & During 1973). Data collected from dairy farms in this study (Table 2) show that the animal losses (via animal products, transfer and preferential flow of urine) for dairy farms on flat topography varied from 3.4 to 4.7 kg K/su. The variation results mainly from differences in management practices between the farms (e.g., the time taken to milk the cows and the use of a feedpad). This variation indicates that the animal losses of K should be calculated independently for each individual farm. The information required is readily available from dairy farmers (e.g., animal production data and the length of time that cows spend off the paddock in the dairy shed).

CONCLUSIONS

A simple mass balance model was used to predict the K losses from three seasonal supply dairy farms. For one of these farms, on a peat soil, it was possible to show that the predictions of soil K losses made by the model were comparable with actual changes in soil K status. Similar evaluations on the other sites were impossible because the changes in soil K with time were insignificant in comparison with the native soil K contents. Although the model is designed to calculate K losses for a typical dairy farm in New Zealand, some modification may be necessary for farms with different management systems. For example, on farms where separate

paddocks are used for day and night grazing, there is usually a transfer of nutrients to the night paddocks which may result in inefficient recycling of K and a loss from the farm.

A comparison made between the losses of K estimated by this model and the model used by the MAF showed several differences between the two. More accurate predictions of the amounts of K lost by the MAF model, and therefore the amounts of K fertiliser required to replace these losses, can be made by considering the loss of K that occurs as a result of the preferential flow of dairy cow urine beyond the plant rooting depth and the length of time that cows spend off the pasture in unproductive areas of the farm.

Fertiliser K requirements predicted by the MAF model are dependent not only on an assessment of animal K losses but also on soil supply factors i.e., MAF Quick Test K values and the estimated K release from K reserves in the soil (the Kc value; Cornforth & Sinclair 1984). Quick Test K measurements in a paddock are highly variable and Kc values are related to New Zealand soil groups irrespective of the land use intensity and mineralogical variation within soil groups. Further improvement of the model will require research into the significance of these soil parameters.

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Appendix 1 Glossary of terms and equations used in the K Loss Model to calculate kg K lost/cow per year.

Terms

M	kg K in milk/cow per year
MF	kg milkfat produced/cow per year
mf	mean milkfat test of the herd for the year (%)
C	kg K in culled cows and calf production/cow per year
W	mean weight of mature cow (kg)
A	mean number of seasons a cow in the herd is milked
c	mean number of calves produced by a cow in the herd in her lifetime
cw	mean calf weight at birth (kg)
P	kg K lost in animal products/cow per year
T	kg K transferred to unproductive areas of the farm/cow per year
h	K concentration in the herbage (% w/w)
t ₁	mean time (h) that the first cow to be milked spends in the dairy shed and raceways a each milking
t ₂	mean time (h) that the last cow to be milked spends in the dairy shed and raceways at each milking
d	length of milking season (days)
f	number of days the herd spends on a feedpad/year
PL	kg K lost due to preferential flow of urine beyond the 15 cm depth of soil/cow per year
plf	proportion of urine that moves by preferential flow beyond the 15 cm depth of soil
TAL	total kg K lost/cow per year

Appendix 1 (*continued*)

Calculation steps

1. $M = (MF \times 0.0013 \times 100)/mf$

[0.0013 represents the K content of milk]

2. $C = [(W/A) \times 0.0015] + [(c \times (A-1)/A) \times cw \times 0.0017]$

[0.0015 and 0.0017 represent the K contents of cow and calf body tissue]

3. $P = M + C$

4. $T = [(MF \times 25 \times h/100) - P] \times [((t_1 + t_2)/24) \times d/365.25] + f/365.25]$

[25 represents the conversion of dry matter consumed to kg milkfat produced; 24 and 365.25 represent the number of h/day and days/year]

5. $PL = [((MF \times 25 \times h/100) - P) - T] \times 0.9 \times plf$

[0.9 represents the proportion of excretal K that is in the form of urine]

6. $TAL = P + T + PL$